Amanda Xia

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EDUCATION

University of Michigan **Ann Arbor, MI**
B.S.E in Electrical Engineering | GPA: 3.87/4.0 *May 2028*
Coursework: Programming and Data Structures, Electronic Circuits, Signals and Systems, Logic Design, Analog Circuits, Computer Organization, Engineering Electromagnetics

SKILLS

Computer Software: Altium, OrCAD Capture/PCB, Onshape, LTspice, Microsoft Office, Cadence Virtuoso, LabVIEW
Electronic Skills: Analyze circuits, Schematic Capture, Component Selection, PCA design process (PCB Layout, SMD soldering, etc.), PCB Bring-Up and Debug (using oscilloscope, multimeter, logic analyzer, power supply, etc.)

EXPERIENCE

Johnson & Johnson MedTech (Ethicon) **Cincinnati, Ohio**
Electrical Engineering R&D Coop *Aug 2026 – Dec 2026*
- Debug analog op-amp filter circuits to determine circuit limitations and areas of improvement
- Developed LabVIEW code with a QMH to automate vessel measurement data acquisition for device reliability testing
- Redesigned and debugged high-power life-test relay circuit in Altium to prevent arcing while maximizing limited space and mechanical limitations that was deployed across 20+ test stations to improve system reliability
- Articulate system requirements including mechanical, electrical, calibration and more to redesign grasping force tester

KLA Corporation **Ann Arbor, MI**
Electrical Engineering Design Intern *June 2026 – Aug 2026*
- Conducted technical evaluation of replacement I²C digital isolators by reviewing datasheets, electrical specifications, and legacy board designs to determine the best-fit solution for system requirements
- Designed a test board using OrCAD to evaluate the robustness and noise reliability of candidate I²C digital isolators
- Created comprehensive technical documentation on component-selection rationale and engineering recommendations
- Presented at weekly meetings, integrating feedback from a cross-functional team to ensure seamless system integration
- Analyzed trade-offs to select and characterized resistors, capacitors, and inductors based on target frequency response, filtering requirements, and circuit functionality

Image Guided Medical Robotics Lab **Ann Arbor, MI**
Undergrad Researcher *Aug 2025 – Present*
- Designed (schematic captured) 4-layer 10mm ring-shaped PCB in Altium with MOSFETS, Infrared LEDs, and Blackfly S synchronized capture system for optical sensing in medical robotics
- Integrated a gaze-tracking subsystem with the larger Robotically Aligned OCT scanner system, implementing compatible power, data, and control interfaces, and designed a CAD case to improve eye identification speed.
- Tested and debugged hardware prototypes (board bring-up) using oscilloscopes, power meters, and multimeters, identifying and resolving issues involving signal integrity, connectivity, and component libraries
- Reviewed and synthesized relevant research papers to inform experimental design and system development

Sensing Intelligence and eXperience Lab **Ann Arbor, MI**
Undergrad Researcher *Aug 2025 – Present*
- Designed and scaled a pressure sensor array circuit from 32×32 to 64×32 by through an expandable signal-routing architecture using multiplexers and digital logic, for higher-resolution sensing without significantly increasing IO pins
- Soldered and debugged 50+ hardware components (MCUs, ICs, passives) by reading through datasheets and using multimeters and microscopes to troubleshoot, ensuring functional prototypes
- Developed and modified Arduino firmware utilizing GPIO, I²C, and SPI peripherals to support evolving hardware

ACTIVITES

Michigan Solar Car Team – *Microsystem Team Member* *Aug 2024 – Present*
Women in Electrical & Computer Engineering (WECE) – *Club Officer* *Aug 2025 – Present*
Girls in Electrical Engineering & Computer Science (GEECS) – *Peer Mentor* *Aug 2024 – Present*
Michigan InVent – *Workshop Facilitator & Instructor* *Aug 2025 – Present*